

Lab 01: Elementary Prog. I

Objective

The purpose of this Lab. is to familiarize students how to solve practical problems programmatically; they will practice on elementary programming using primitive data types, variables, constants, operators, expressions, and input and output. Also, they will learn how to diagnose errors that may occur when a program is compiled or executed. There are some exercises, through which they will understand the concepts they learnt in this chapter.

Current Lab Learning Outcomes (LLO)

By completion of the lab the students should be able to

1. Use of Scanner class to obtain input from the console.
2. Define and declare identifiers, naming variables, constants, methods, and classes.
3. The use of constants, Java primitive data types: byte, short, int, long, float, double, and char
4. Familiarity with Java operators to write numeric expressions, use shorthand will be taught to them.
5. Able to distinguish syntax errors, runtime errors, and logic errors and debug errors

Lab Requirements

NetBeans

Lab Description

Theory Review (5 minutes)

Java provides simple data types for integers numbers (byte, short, int, long), real numbers (float, double), characters, string and boolean types.

In order to use the values(data) in the program, we need to use variables which are used to represent values stored in the computer's memory.

The variables used in the program must be declared (determining the type and name).

The syntax for variables declarations is:

type var_name; where **type** is any data type mentioned above and **var_name** is the variable name.

Example:

double radius;

the variable radius will store the data of real numbers;

The variable must be declared before it can be assigned a value, and a value must be assigned to the variable before it can be used.

To assign a value to the variable, follow this syntax:

var_name = value;

Example:

radius = 10.5;

The variable declaration and assignment statement can be in a single statement as:

type var_name= value;

Example:

double radius = 10.5;

Sometimes, we need to use a constant value in the program.

To do this, follow the syntax given below:

final type const_name = value;

Example:

```
final double pi = 3.14;
```

Practice Activity with Lab Instructor (10 minutes)

Write a program in java to interchange (swap) values of two variables with a use of 3rd variable.

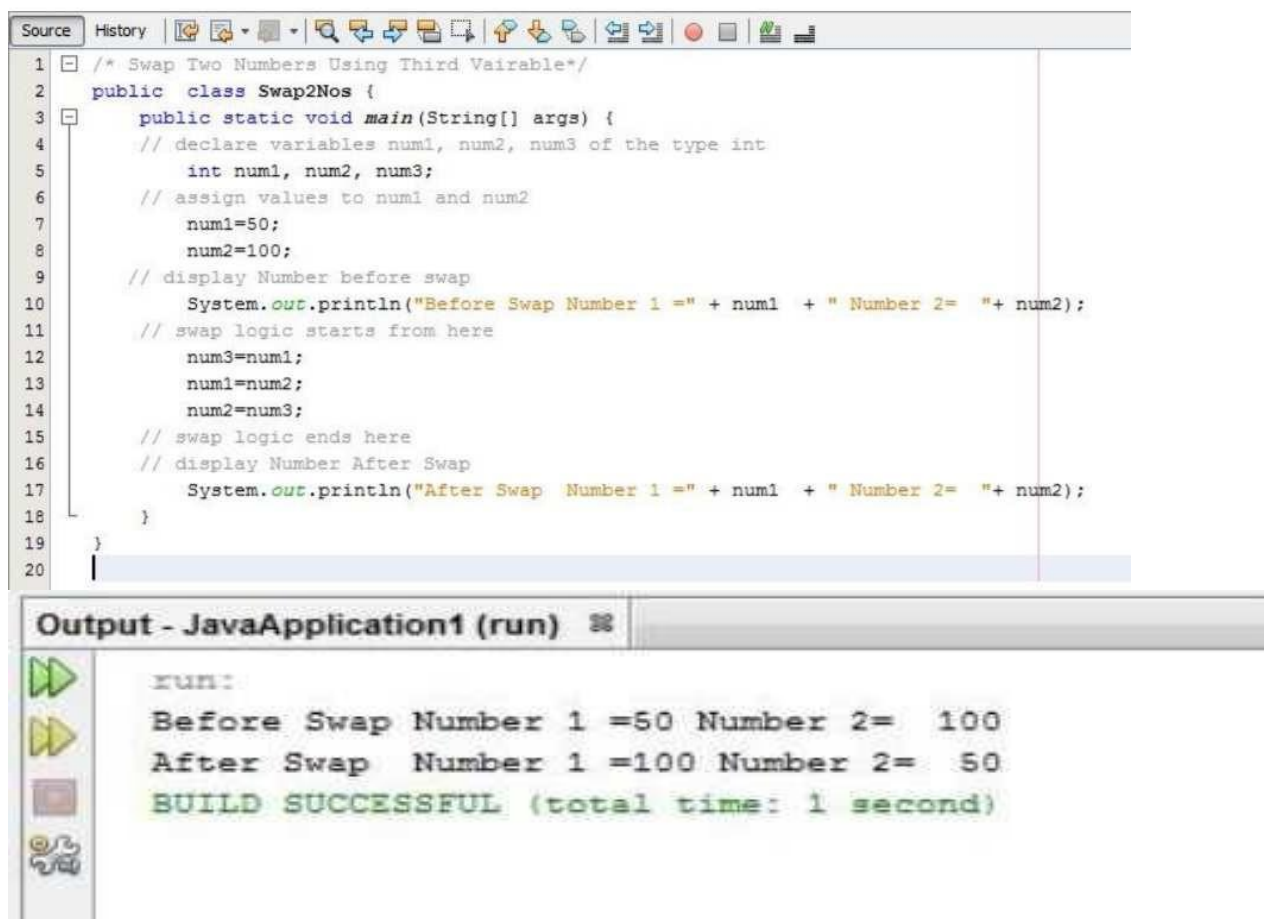
Input: Value of num1 and num2

Processing: swapping values of num1 to num2 and num2 to num1 using a third variable num3.

Output: Display value of num1 and num2 after swap operation.

Solution:

1. Open NetBeans and create a new project
2. Create a new java main class and write its name as Swap2Nos
3. Write the following code inside the main method



```
1  /* Swap Two Numbers Using Third Vairable*/
2  public class Swap2Nos {
3      public static void main(String[] args) {
4          // declare variables num1, num2, num3 of the type int
5          int num1, num2, num3;
6          // assign values to num1 and num2
7          num1=50;
8          num2=100;
9          // display Number before swap
10         System.out.println("Before Swap Number 1 =" + num1 + " Number 2= " + num2);
11         // swap logic starts from here
12         num3=num1;
13         num1=num2;
14         num2=num3;
15         // swap logic ends here
16         // display Number After Swap
17         System.out.println("After Swap Number 1 =" + num1 + " Number 2= " + num2);
18     }
19 }
20
```

Output - JavaApplication1 (run)

```
run:
Before Swap Number 1 =50 Number 2= 100
After Swap Number 1 =100 Number 2= 50
BUILD SUCCESSFUL (total time: 1 second)
```

Lab Assessment

Three Problems\ exercises should be completed **in 60 minutes individually**

1. Write a program that converts pounds into kilograms. The program should ask the user to enter a number in pounds. The program should then convert this number into kilograms and then display the result. *Note: one pound is 0.454 kilograms.

Here is a sample run of the program:

```
Enter a number in pounds: 55.5 
55.5 pounds is 25.197 kilograms.
```

2. Tipping for service is very common in many countries, and the normal percentage for tipping is 15%. For example, if you eat at a restaurant and the total bill comes to 50, the normal tip would be 15% of the 50, which is 7.5. So the total you pay will be 57.5. But perhaps in another country, 10% is the normal tipping percentage.

In this case, the tip would be 5, for a total of 55. Write a program that reads the subtotal and the desired tip rate. Your program should then compute the tip, the new total, and should display the results. Here is a sample run of

the program:

```
Enter the subtotal: 50 
Enter the tip rate: 15 
The tip is 7.5 and the total is 57.5sr.
```

Here is one more sample run:

```
Enter the subtotal: 78 
Enter the tip rate: 5 
The tip is 3.9 and the total is 81.9sr.
```

3. Write a program that asks the user to enter the number of minutes (e.g. 1 billion). Your program should then display the number of years and days from those minutes. *You can assume that a year has exactly 365 days. Here is a sample run of the program:

```
Enter the number of minutes: 1000000000 
1000000000 minutes is approximately 1902 years and 214 days.
```

The program above is used to swap number 50 with number 100. If we want to swap another values we must go to the program and change the values of variables **num1** and **num2**. To make a program general i.e. to swap any two values without needing any change in the program, we need to enter the values from the keyboard. So, to do this we will use a Scanner class.

The following statements are used to input the values from the keyboard.

- `Scanner scanner_name= new Scanner(System.in);`
- `System.out.print("Any Suitable Message: ");`
- `var_name= scanner_name.input_method;`
- The character with red color should be capital letter.
- `scanner_name` : any identifier name
- **Any Suitable Message**: any understandable message for user to enter the value from the keyboard (for example "Enter the name: " for entering the name or "Enter the Radius of Circle: " to enter the value of radius and so on).
- `input_method`:
 - `next()`: if the `var_name` is of type **String**
 - `nextInt()`: if the `var_name` is of type **int**
 - `nextDouble()`: if the `var_name` is of type **double**
 - `nextFloat()`: if the `var_name` is of type **float**
 - `nextByte()`: if the `var_name` is of type **byte**
- `Scanner scanner_name= new Scanner(System.in);` is written only one time in the program. Before using this statement, we need to write the statement `import java.util.Scanner;` before the class name and at the beginning of file.

Short Exercise (5 minutes)

Modify the above program by using Scanner, and then ask the user to enter values of num1 and num2 from the keyboard.

*Hint: use one Scanner variable and read the values of num1 and num2 from the user. Everything else is the same. In summary, you only change three lines!

The screenshot displays an IDE with a Java source file. The code defines a class `Swap2Nos` with a `main` method. It uses `java.util.Scanner` to take input from the user. The code prints the numbers before and after a swap operation. The output window shows the execution results, including the input values (100 and 200) and the final state of the numbers after swapping (200 and 100).

```
1
2 import java.util.Scanner;
3 /* Swap Two Numbers Using Third Variable */
4 public class Swap2Nos {
5     public static void main(String[] args) {
6         Scanner input=new Scanner(System.in);
7         // declare variables num1, num2, num3 of the type int
8         int num1, num2, num3;
9         // Enter values to num1 and num2 from the keyboard
10        System.out.print("Enter values of first and second numbers: ");
11        num1=input.nextInt();
12        num2=input.nextInt();
13        // display Number before swap
14        System.out.println("Before Swap Number 1 =" + num1 + " Number 2= " + num2);
15        // swap logic starts from here
16        num3=num1;
17        num1=num2;
18        num2=num3;
19        // swap logic ends here
20        // display Number After Swap
21        System.out.println("After Swap Number 1 =" + num1 + " Number 2= " + num2);
22    }
23 }
```

Output - JavaApplication1 (run)

```
run:
Enter values of first and second numbers: 100
200
Before Swap Number 1 =100 Number 2= 200
After Swap Number 1 =200 Number 2= 100
BUILD SUCCESSFUL (total time: 12 seconds)
```